Can the percent of phospholipase A2 in the total protein makeup of cobra venom be estimated from RP-HPLC-UV peak areas, and can this method track PLA2% to the ability to spit venom, rather than to being a true cobra (genus Naja)?

RP-HPLC-UV chromatograms of whole venoms for five species: two non-spitting true cobras (*Naja naja, Naja haje*), two spitting true cobras (*Naja pallida, Naja nigricollis*), and one spitting "cobra" outside of the Naja genus *(Hemachatus haemachatus*), will be used. This project asks if absorbance area, corrected for relative extinction coefficients, can be used to estimate PLA2 levels. Extinction coefficients will be estimated per species using published relative abundances of proteins in the two primary categories of toxin in these snakes.

Data

- Arrahman et al. (2022), RP-HPLC-UV Chromatograms, .csv, https://doi.org/10.6084/m9.figshare.19447583.v3, accessed 11/07/2026. One csv per sample, one sample per species, run without replicates, reported as retention time (10-30 min, 0.3 s steps) and signal at 220 and 254 nm (units not stated, they cancel in the percent of total). The paper assigns PLA2 retention times for each venom. The two other files are mambas and are not used.
- Kazandjian thesis (2020), toxin sequences and abundances, .xlsx, https://datacat.liverpool.ac.uk/1186/, accessed 11/07/2026. One .xlsx per species, one sample per species combining venom from 2-6 snakes, analyzed without replicates using top-down proteomics, reported as %TIC for each toxin detected. Used only to weight absorptivities. 12 additional cobra species were measured and are not used here.
- Nguyen et al. (2022), toxin types and abundances, .xlsx, https://doi.org/10.5524/102333, accessed 11/08/2026. One .xlsx for all species, one sample per species pooled from unknown number of snakes, analyzed without replicates using bottom-up proteomics, reported as %mol for each toxin category detected. Used only as a check against assumptions made. Only four of the five species looked at here are included and 19 additional species were measured and are not used.

Methodology

After baseline correction of each chromatogram, I will integrate the PLA2 peak(s) at the retention times assigned in the paper and divide by the total area at 220 nm. The assumption that toxins outside of the PLA2 family are in the 3FTx family will be made, based on publish reports that these are the two primary toxin families for elapid snakes, which these all are, and that other types of toxins are in comparatively very low abundance for these snakes (Kazandjian, 2020). I will convert area to a mol fraction with a PLA2:3FTx absorptivity ratio calculated per species, from amino acid sequences (residue and peptide-bond values at 214 nm, Kuipers & Gruppen 2007) averaging over all variants in each toxin family, weighted by abundance of variants in each species. The weighting matters because within a given toxin family, particularly 3FTx, there can be sub-types of proteins that vary in size significantly, and often different snakes will have different abundances of these sub-types, shifting the molar absorptivity enough to differ from the unweighted average. The check will be a comparison with bottom-up proteomic PLA2 abundances from Nguyen et al, which cover four of the five species (*N. naja* is not included).

Data sufficiency

- Sample size: Five venoms, one sample each, only one outside of *Naja*. The data can show whether the method agrees with proteomics for these species, not that it works across all cobras or detects variation within species.
- Peak assignments and baseline: PLA2 peaks are indentified but retention times differ from the paper's by 0-0.15 min, so I will assign a delay per chromatogram and confirm the validity of this assumption by checking the alignment in spectra where two PLA2 peaks are reported. Some spectra have a stronger shifting baseline than others, so this will also be corrected per chromatogram.
- Two-component assumption: Non-PLA2 peaks are not identified, and other families elute near the 3FTxs, so differentiation of 3FTx and non 3FTx within the non-PLA2 peaks is not possible. This assumes other components are nearly absent from venom, which is weakest for *H. haemachatus* which has ~5-10% other toxins. These proteins are also nearly absent from the top-down data, which may miss large proteins such as these, so this is only reflected in the check.
- Concentration from UV data: absorption is not reported in AU, but this cancels in the % calculation. Absorptivities are calculated from 214 nm values, but the data is at 220 nm, increasing uncertainty.
- Species label: the chromatogram file for *N. nigricollis* is named NAJMO, the typical abbreviation for *N. mossambica*. This file might be mislabeled, or they may have used *N. mossambica* instead because prior to DNA sequencing, this was believed to be a sub-species of *N. nigricollis*. Both species appear in the Kazandjian and Nguyen data, so the results can be compared with each.
- If the data cannot answer the question because a PLA2 peak cannot be resolved, I will fit overlapping peaks and report a range. If because estimates disagree with proteomics beyond the uncertainty, I will test if any of my assumptions accounts for this.